

Continuous Treatment Difference-in-Differences with Unknown Controls: A Data-Driven Approach

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Abstract

This paper studies difference-in-differences (DD) research designs where all observations receive a continuous treatment (or dose) in response to an aggregate policy, so there is no group that is *ex post* unexposed. This setting stands in contrast to the recent literature re-examining DD estimators which typically requires that a subset of observations never receive the treatment to identify the Average Treatment Effect on the Treated (ATT). We develop a framework to estimate the ATT when the dose takes effect only after a cutoff value, the Minimum Effective Dose (MED), and propose a sample splitting estimator of the ATT and MED under non-parametric assumptions on the dose response function: first estimate the cutoff in a hold-out sample, then use it to define and estimate the ATT with least squares. This estimator is consistent and the second-stage standard error is estimated with usual methods to conduct inference. We conclude with simulations comparing performance to current methods.

JEL codes: C14, C23, C24.

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1 Introduction

Without additional assumptions, it is generally impossible to infer the effects of a treatment by comparing participants to non-participants or comparing the participants over time. For valid inference, unit comparisons must restrict selection into treatment, which is generally difficult to justify in the absence of an instrument. On the other hand, time comparisons (e.g. interrupted time series methods) cannot allow for contemporaneous trends. Difference-in-differences (DD) research designs combine these two estimators to infer causal effects by subtracting the change in outcomes over time for non-participants from the change for participants. This is valid whenever the average change in outcomes for participants in the absence of treatment (a counterfactual moment) is equal to the average change in outcomes for those who go untreated, known as a Parallel Trends assumption (PTA). Panel data thus enables identification of a causal effect without ruling out selection into treatment levels or contemporaneous trends.

A common extension in practice admits a continuously distributed treatment (or dose) variable, where all units are exposed to some level of the dose $D_i \geq 0$. A prominent example of this method is [Card \(1992\)](#) which measures a state's exposure to a higher federal minimum wage by the share of teenage workers who fall below the new legislated minimum.¹ The estimating equation follows via analogy to binary difference-in-differences

$$Y_{i,t} = \alpha_i + \theta_t + \beta \cdot D_i P_t + \epsilon_{i,t} \quad (1)$$

where $Y_{i,t}$ is some outcome of interest, α_i and θ_t are unit and time fixed-effects respectively, P_t indicates the periods that units are exposed, and ϵ_{it} is some error term. The main coefficient of interest in the Two-Way Fixed-Effects (TWFE) regression above is β , which represents an aggrega-

¹A common structure behind many continuous treatment difference-in-differences research designs is an aggregate policy where the researcher hypothesizes that units face heterogeneous exposure according to some dose variable.

tion of the “effect” of the continuous treatment. Researchers have typically interpreted this as the weighted average dose response function (DRF), in line with the “average derivative” interpretation of regression coefficients (Yitzhaki, 1996; Angrist and Pischke, 2009).

Recently Callaway et al. (2024), hereafter referred to as CGS, decompose β and derive sufficient assumptions to identify some well-defined causal parameter. They show that whenever researchers have access to “pure control” units (defined as those which are *ex ante* unexposed to the treatment with dose $D_i = 0$) only minor modifications to PTA identify the Average Treatment Effect on the Treated (ATT) parameter, defined for each positive dose level.

But what about cases when researchers do not have access to these units? To understand the data constraints that researchers face, we conducted a metastudy of all papers published in the *American Economic Review* (AER) between 2000 and 2018. Out of the 44 papers we conclude estimates a continuous treatment DD model, only 19 have access to a pool of untreated units. We explicitly consider a setting where no units are untreated to better understand inference in these settings.

We find that 70% of the papers in our metastudy estimate a full dose regression like (1). Without pure control units, this strategy relies exclusively on comparisons across doses, aggregating these local comparisons into an estimate of the overall dose response. However, the standard parallel trends assumption does not identify the ATT, nor the average effect of marginal increases in the dose, or Average Causal Response (ACR). Intuitively, this is because comparisons across treated units at different doses must additionally restrict selection into treatment levels, which is not typical in standard DD settings. In the case of a continuous dose, the least squares estimator of β in (1) converges to a weighted average of differences between adjacent ATT estimates across the dose distribution, which only reveals a causal response under strong restrictions on counterfactual heterogeneity (what CGS call strong parallel trends).

If we are not willing to make strong assumptions, what is left to do? The remaining 30% of papers dichotomize the dose variable and estimate a traditional difference-in-differences model. This approach bins all “intensely” exposed units together and compares them with everyone else; the hope is that even if some control units receive a small dose, as long as the effect is monotonic this will reveal an attenuated version of the ATT. This approach is not cost free, however, as it throws away all dose information and potentially useful comparisons.

We propose using this variation to shed light on estimators which dichotomize a continuous treatment to recover a difference-in-difference design. In Section 2 of the paper, we show that a continuous TWFE design can be thought of as an above-mean/below-mean comparison, which implies our results apply more broadly to inference in settings where no pure controls are available.

We begin by constructing a potential outcomes framework for applications where all units are treated at a certain time period, and consider a counterfactual where all units are untreated which allows us to formalize precise definitions of treatment effect parameters of interest. Inference in these settings is particularly difficult, beyond the usual Fundamental Problem of Causal Inference ([Imbens and Rubin, 2015](#)) that treated and untreated outcomes cannot be observed for the same unit at the same time. In this case, in every period, we observe *either* a treated or untreated outcome for *all* units, so that contemporaneous treatment-control comparisons cannot be made. This issue is well known in the macroeconomics literature ([Rambachan and Shephard, 2019](#)), which exploits the timing of exogenous shocks for identification. Instead, we focus on the use of treatment-control comparisons, which remains the dominant paradigm in applied microeconometrics.

Next, we derive a data-driven, non-parametric estimator of the average treated effect on the treated in a setting where all units are *ex ante* treated; that is, a setting where no zero dose units are observed. This approach hinges on the assumption that certain groups might experience outcomes

identical to a world in which the policy is not passed. We call this a “Minimum Effective Dose” (MED), following the pharmaceutical literature, where units receiving a dose lower than the MED experience an outcome indistinguishable from a zero dose control group (see [Ruberg \(1989\)](#) for a formal definition). We propose a two-step non-parametric approach to identifying the ATT when a MED exists. In the first step, using a p -value approach from [Mallik et al. \(2011\)](#), we estimate the MED; then, in the second step, we utilize this parameter to estimate the ATT. To avoid bias in inference on the ATT from simultaneous model selection, we recommend a sample-splitting approach where the dataset is split into two parts to tackle each of these tasks.

Further, we develop a statistical test of the hypothesis that there is no dose response below the estimated MED. This is analogous to a pre-trends test of the parallel trends assumption, and is essential for unbiased estimation of the ATT in this setting. The split sample estimator enables a direct way to implement this falsification test, by focusing on the second group used to estimate the ATT. Since the variation within groups is not used for estimation, we can isolate only those units classified as controls to implement the test.

Finally, we present a number of important corollaries and extensions to our main results. In the case where researchers are willing to make parametric assumptions, existing methods can be used to estimate the ATT in a more efficient way. This is accomplished by either extrapolating estimated pre-trends with at least one additional period of data, or by specifying a functional form for the dose response function and estimating it jointly with the MED in one stage. We then address the case where the MED cannot be assumed to exist, and show that the ATT is fundamentally unidentified. This implies researchers must compromise and target an alternative estimand, which we call the Binned Average Treatment Effect on the Treated (BATT), an average of individual treatment effects across a flexible range of the dose.

1.1 Connections to Literature

This paper contributes to three literatures. Recent technical advances to difference-in-differences dealing with staggered adoption ([Goodman-Bacon \(2021\)](#); [Sun and Abraham \(2021\)](#); [Wooldridge \(2021\)](#); [de Chaisemartin and D’Haultfoeuille \(2020\)](#)), heterogeneous treatment effects ([Goodman-Bacon \(2021\)](#); [Callaway and Sant’Anna \(2021\)](#); pre-testing ([Roth \(2022\)](#)), and functional form specification ([Roth and Sant’Anna \(2021\)](#)) are increasingly well understood. However, identification issues related to continuous treatments are actively being litigated among econometricians and applied researchers alike ([Callaway et al. \(2024\)](#); [de Chaisemartin et al. \(2023\)](#); [Sun and Shapiro \(2022\)](#); [Butts \(2022\)](#)). Our main contribution is to show that researchers can still rely on traditional parallel trends assumptions even with continuous treatments that lack pure controls and to clarify the assumptions that this type of inference requires. This is not a free lunch though, since our estimator results in a loss of power when estimating the ATT due to its reliance on a smaller subsample based on the generated MED regressor. We must also assume that an MED exists, which is not always guaranteed in practice. In Section 3 we present a model of dynamic constrained investment which gives rise to a zero-response region, motivating the MED existence assumption.

The approach in [Butts \(2022\)](#) to estimate treatment effects at specific locations with geocoded data is most similar to ours. In studying treatment explicitly as continuous distance, he notes that researchers must know the threshold distance beyond (or below) which treatment effects begin, and proposes a non-parametric method to estimate the treatment effect curve (what we call the dose response function) with large data. Similarly, existing approaches to identifying treatment effects in a difference-in-differences setting rely on additional variation not present in our set-up. Our framework is inspired in large part by [Callaway et al. \(2024\)](#), but they assume that *ex ante* untreated units are available and observed by the researcher. [de Chaisemartin et al. \(2023\)](#) iden-

tify causal responses with additional variation in the treatment variable: If the dose changes over time and there exists a group of stayers or a group of quasi-stayers, then the average derivative among the treated is identified. [Sun and Shapiro \(2022\)](#) provide impossibility results on identifying causal responses in TWFE regressions absent such additional information. They show that with the existence of a pure control, a modified instrumented difference-in-differences approach is sufficient to target an average of causal responses among treated units. We contribute to this work by providing a consistent framework to identify and estimate the ATT without pure controls, a case that is common in practice.

This paper leverages many findings from the literature on threshold and change point estimation. In a simple model of a minimum effective dose with a homogenous linear dose response, the MED constitutes a changepoint in an underlying linear model. Inference in this setting is complicated because the changepoint is not identified if the null hypothesis is true ([Hansen, 1996](#))—that is, in our simple example, if there is no dose response. We show how these methods can be utilized in a continuous difference-in-differences setting whenever relevant parametric assumptions hold. [Hansen \(1999\)](#) develops this theory in a panel data setting, which now even has a dedicated Stata command ([Wang, 2015](#)). Recent work has developed theory to accommodate dynamic panels ([Seo and Shin, 2016](#)), interactive fixed effects ([Miao et al., 2020](#)), and heterogeneous breakpoint and slope coefficients ([Miao et al., 2020](#)).

Finally, we contribute to a literature spanning several disciplines which highlights the problems inherent to dichotomizing a continuous variable. The motivation for this specification varies by application: the psychology literature dichotomizes either one or two continuous variables to use a more familiar one-way or two-way ANOVA design ([MacCallum et al., 2002](#)); the clinical literature targets a threshold for biomarkers that maximizes the predictive use of flagging patients who fall above this threshold ([Altman et al., 1994](#)); and the epidemiology literature categorizes continuous

controls into several groups to model a more flexible linear specification² (Brenner, 1997). This literature may not be familiar or relevant to many economists, who usually do not shy away from exploiting continuous variation in data.³ However, the lure of a difference-in-differences design⁴ has led to the emergence of this practice in the applied microeconomics literature. We argue that leads to similar problems noted in earlier work, and warrants a clear statement of necessary identifying assumptions that must hold to identify meaningful estimands. We aim to do exactly this in the remainder of the paper.

1.2 Organization of the Paper

Section 2 describes our potential outcomes framework and formalizes the baseline assumptions needed for inference. In Section 3, we propose our non-parametric estimator and statistical tests of the MED existence assumption. Section 4 concludes.

2 Framework

2.1 Setup

Following Callaway et al. (2024), we consider an environment with two time periods $t \in \{\tau - 1, \tau\}$ and N units $i \in \{1, \dots, N\}$, where some outcome $Y_{i,t}$ is observed. Units are assigned a continuously distributed treatment D_i , which is also observed by the researcher. Data are independent and identically distributed across units:

A1 Random sampling: $\{Y_{i,\tau}, Y_{i,\tau-1}, D_i\}_{i=1}^N$ is independent and identically distributed (iid)

²For example, instead of using the number of cigarettes smoked per week, this variable might be dichotomized into “non-smoker,” “light smoker” and “heavy smoker.”

³In fact, widely known work on measurement error would immediately identify this as problematic (see Bound et al. (2001) for a summary).

⁴In particular, the ability to falsify its main identifying assumption (parallel trends) with a visual and statistical pre-trends test given the availability of at least one additional period before treatment.

The dose distribution is given by the cumulative distribution function $F_D(d)$ with bounded and compact support on $\mathbb{R}_+$ and a well-defined probability distribution function. Formally:

A2 Dose distribution: $D_i \sim F_D(d)$ over compact $\text{supp}\{D_i\} := \mathcal{D} \subset \mathbb{R}_+$, which admits a Radon-Nikodym derivative $f(d)$ such that $f(d) > 0 \forall d \in \mathcal{D}$. Let $d_l = \inf \mathcal{D}$ and $d_u = \sup \mathcal{D}$.

In most applications, the support $\mathcal{D}$ is either discrete or continuous over a closed interval. To facilitate discussion, we clarify assumptions for each of these cases:

A2.1 Discrete dose distribution: The support of the dose distribution is given by $\mathcal{D} = \{d_1, \dots, d_J\}$ for some finite J where $0 < d_1 < d_2 < \dots < d_J$. The CDF $F_D(d)$ admits a probability *mass* function $f(d)$ such that $f(d) > 0 \forall d \in \mathcal{D}$.

A2.2 Continuous dose distribution: The support of the dose distribution is given by $\mathcal{D} = [d_l, d_u]$ for some $0 < d_l < d_u$. The CDF $F_D(d)$ admits a probability *density* function $f(d)$ such that $f(d) > 0 \forall d \in \mathcal{D}$.

2.2 Potential Outcomes

In each period, let $Y_{i,t}(D_i, L_t)$ denote a bivariate potential outcome. The first argument is the dose assigned to each unit. We assume some policy of interest to the researcher is implemented at time τ . The second argument, L_t , is an indicator for the policy of interest being implemented in time τ . In all applications of interest, $L_{\tau-1} = 0$ and $L_\tau = 1$, but the purpose of this variable is to define a counterfactual where the policy was not implemented ($L_\tau = 0$) for all units.

Using this bivariate structure, we place two restrictions on observed outcomes. First, we make the standard assumption that the policy is not anticipated so that observed outcomes in period $\tau - 1$ are equal to untreated potential outcomes. Formally,

A3.1 No anticipation: $Y_{i,\tau-1} = Y_{i,\tau-1}(D_i, 0)$

We also assume the dose is policy-relevant, meaning it only impacts potential outcomes if the policy is enacted. Formally,

A3.2 Policy-relevance: For $d, d' \in \mathcal{D}$, $Y_{i,t}(d, 0) = Y_{i,t}(d', 0)$ for $t \in \{\tau - 1, \tau\}$

With $P_t = 1\{t = \tau\}$ indicating the time when the policy is implemented, observed outcomes are

$$Y_{i,t} = (1 - P_t)Y_{i,t}(D_i, 0) + P_t Y_{i,t}(D_i, 1) \quad (2)$$

Although the policy indicator L_t is not used for estimation, it enables us to consider a useful thought experiment, which we use to define our target estimands. Our building block is an *individual treatment effect* (ITE), or the difference between the observed (treated) outcome and a counterfactual one in a world where the policy was not implemented:

$$\mu_i(D_i) \equiv Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) \quad \text{ITE}$$

By considering how this changes as the dose is varied we trace out the dose response function (DRF), which might vary across units. We target a variety of averages of these parameters. Following [Callaway et al. \(2024\)](#), the *average treatment effect at dose d* (ATT-d) is given by

$$\mu(a|d) \equiv \mathbb{E}[Y_{i,\tau}(a, 1) - Y_{i,\tau}(a, 0) \mid D_i = d] \quad \text{ATT-d}$$

Intuitively, the individual treatment effect of receiving dose a is averaged over all units receiving the dose d . Of particular interest is the average effect at the actual dose received, $\mu(d|d)$. We can generalize this idea to define the *binned average treatment effect on the treated* (BATT), which averages

individual treatment effects over a range of doses instead of a single dose:

$$\text{BATT}[d, d'] \equiv \mathbb{E}[Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) \mid D_i \in [d, d']] \quad \text{BATT}$$

While this terminology is new, the concept is not; we are simply extending the notion of a dose-specific average treatment effect on the treated (ATT-d) utilized in [Callaway et al. \(2024\)](#). Finally, letting $T_i = 1$ denote all units that receive treatment allows us to define the *average treatment effect on the treated* (ATT):

$$\mathbb{E}[Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) \mid T_i = 1] \quad \text{ATT}$$

Note that even if all units are treated, this is still well-defined. The ATT is our primary estimand of interest, framing our discussion of identification.

2.3 Current Practice

Before describing our proposed non-parametric method, we use the framework described above to characterize current practice in more detail. In particular, we focus on specifications which dichotomizes the continuous treatment at some researcher-specified threshold. One reason for this approach is to identify causal effects under minimal assumptions on the data-generating process, which we also pursue by eschewing parametric assumptions.

2.3.1 Metastudy

To get a sense of the use and justification of a binned approach, we conduct a small metastudy. Using Google Scholar, we query all papers published in the *American Economic Review* between 2000 and 2018 which contain the keywords “difference-in-difference” and “continuous” in the

manuscript. We find 178 total papers that satisfy these requirements and after checking each one, retain only those that estimate a continuous treatment difference-in-difference model. As we state in the introduction, 70% of papers estimate a full dose regression like (1). The remaining 30% of papers dichotomize the dose at a given value or percentile, and compare units above and below some researcher-defined cutoff. In our framework, this is equivalent to the bivariate regression:

$$\Delta Y_{i,t} = \alpha + \beta \cdot \mathbb{1}(D_i \geq d_r) + \varepsilon_{i,t} \quad (3)$$

Researchers choose some threshold d_r and use more exposed units ($d_i \geq d_r$) as a “treatment” group to compare to less exposed units ($d_i < d_r$) in a “control” group, to recover a binary DD structure. Our metastudy suggests that one prominent motivation to dichotomize is to recover a traditional difference-in-differences setup that relies on comparisons between treated and control units. In the “2” $\times$ 2 setting here, regression recovers a classic difference-in-differences of means:

$$\hat{\beta} = \hat{\mathbb{E}}[\Delta Y_{it} | D_i \geq d_r] - \hat{\mathbb{E}}[\Delta Y_{it} | D_i < d_r]$$

Where $\hat{\mathbb{E}}$ denotes the sample average equivalent of the population expectation. While this design has intuitive appeal, it is unclear what the causal estimand is in this approach, let alone if the binned estimator is unbiased or consistent.

2.3.2 What Parameter Does TWFE Recover?

Consistent with this design mimicking a standard binary difference-in-differences design, practitioners often invoke a standard parallel trends assumption. We introduce this assumption formally and consider what this design will estimate:

$$\text{A4 Parallel Trends: } \hat{\mathbb{E}}[Y_{it}(D_i, 0) | D_i = d] = \hat{\mathbb{E}}[Y_{it}(D_i, 0) | D_i = d'] \quad \forall d, d' \in \mathcal{D}$$

Intuitively, since the “control” group in this setting receives a treatment, we do not recover the ATT, but rather the difference in average treatment effects for both groups:

Proposition 1. *Suppose that assumptions A1-A4 hold. Then, the binned difference-in-differences estimator will recover $BATT[d_r, d_u] - BATT[d_l, d_r]$.*

Note that, even though untreated outcomes are not observed in period τ , we arrive at an expression involving causal estimands as untreated potential outcomes are differenced out. However, the cost of this is that we are only able to estimate the difference in BATT parameters.

2.3.3 Bias in ATT Estimation

There is no clear relationship between the term in Proposition 1 and the ATT. In fact, they are in tension; while the binned estimator recovers the difference of BATTs for the “treated” and “control” group, the ATT is the weighted sum of these objects:

$$ATT = F(d_r)BATT[d_r, d_u] + (1 - F(d_r))BATT[d_l, d_r]$$

Perhaps the clearest way this can cause problems is if the dose response function is not monotonic.

Note that if $BATT[d_l, d_r] > BATT[d_r, d_u] > 0$, it will be the case that $ATT > 0$ but $\mathbb{E}[\hat{\beta}] < 0$.

To avoid this outcome, we would need to impose the further assumption that $BATT[d_r, d_u] > BATT[d_l, d_r]$. However, since the choice of d_r is arbitrary, we would functionally need to assume that $BATT[d_r, d_u] > BATT[d_l, d_r] \forall d_r \in (d_l, d_u)$. This would ensure that $\mathbb{E}[\hat{\beta}]$ has the same sign as the ATT, but the bias from such an estimator would still have unknown sign.

To see this, we can show that the bias of the binned estimator can be written as:

$$\mathbb{E}[\hat{\beta}] - ATT = F(d_r)BATT[d_r, d_u] - (1 - F(d_r))BATT[d_l, d_r]$$

As this expression illustrates, bias from this approach depends not only on the differences in BATTs but also on dose distribution through the choice of cutoff d_r .

In general, it is not possible to “rescue” this approach by making parametric assumptions. To see this, consider the case of a homogenous, linear dose response function, given by $\mu(d_i) = \beta d_i$, with a dose distribution that is approximately normally distributed with mean μ and standard deviation σ .⁵ The *ATT* in this case will simply be equal to $\beta\mu$. However, using well-known results from selection models, we show in the appendix that the binned estimator will yield

$$\beta\sigma \frac{\phi(\frac{d_r - \mu}{\sigma})}{\Phi(\frac{d_r - \mu}{\sigma})(1 - \Phi(\frac{d_r - \mu}{\sigma}))}$$

Which is approximately equal to $\beta\sigma$ when d_r is chosen to be the median. In this case, the binned estimator is a function of the dose distribution’s standard deviation, while the *ATT* is a function of the dose distribution’s mean, even under the restrictive assumption of a linear dose response.⁶

2.3.4 Inflated Type 1 Error Rate

The analysis thus far implicitly assumed that the researcher cutoff d_r was chosen without regard to what the data look like. However, since there is not much consensus on how to choose a cutoff for empirical analyses, this is largely left to researcher discretion. This is worrying as the strategic choice of where this cutoff is can lead to substantial inflation of the Type I error rate. This is very well known in the clinical literature, where it was common practice to choose a cutoff for some biomarker by minimizing the p -value, arguing that such a procedure would lead to the most predictive cutoff point (Altman et al., 1994). While we are not aware of any work promoting this approach, concerns about p -hacking and publication bias in difference-in-differences (Brodeur

⁵This is approximate as any normal distribution can take negative values, violating A2.

⁶We provide a rationalization of this approach when targeting $BATT[d_r, d_u]$ in the appendix.

et al., 2020) suggest that this should be a concern.

It will come as no surprise that choosing a threshold “optimally” - that is, with the lowest p -value - will lead to an inflated Type I error rate. Rather, it is the degree of inflation that occurs that is of concern, which can be calculated (approximately) from a known asymptotic distribution.

Recall the model in 3 where the researcher regresses an outcome ΔY_i on a binarized dose variable. We know that the coefficient estimate $\hat{\beta}$ will converge to $\mathbb{E}[\Delta Y_i | D_i \geq d_r] - \mathbb{E}[\Delta Y_i | D_i < d_r]$ as this is simply a t -test of the difference in means between the “treatment” and “control” groups. Let $T_n(d)$ denote the value of this t -statistic for sample size n and cutoff d . We consider the maximum of these statistics from some lower bound d_1 to some upper bound d_2 , denoted by

$$\underset{d \in [d_1, d_2]}{\text{maximize}} \quad |T_n(d)| \quad (4)$$

Lausen and Schumacher (1992) show that this object converges to the supremum of the absolute value of a standardized Brownian bridge, given by

$$\sup_{t \in [\epsilon_1, \epsilon_2]} \frac{|B_0(t)|}{(t(1-t))^{1/2}} \quad (5)$$

Where $\epsilon_1 = F(d_1)$ and $\epsilon_2 = F(d_2)$. Miller and Siegmund (1982) provide the following asymptotic approximation for the Type I error of this distribution:

$$\mathbb{P} \left[\sup_{t \in [\epsilon, 1-\epsilon]} \frac{|B_0(t)|}{(t(1-t))^{1/2}} \geq z \right] = \frac{4\phi(z)}{z} + \phi(z) \left(z - \frac{1}{z} \right) \ln \left(\frac{(1-\epsilon)^2}{\epsilon^2} \right) + o \left(\frac{\phi(z)}{z} \right) \quad (6)$$

Where we have simplified the set of cutoffs to search over to the symmetric range $[\epsilon, 1 - \epsilon]$. Using the z -score that would be used to define the rejection region of this test, we can calculate the actual Type 1 error probability would result. We reproduce part of Table 1 from Miller and Siegmund

(1982) to give a sense of how these levels change.

Table 1: Inflation Rejection Rates of Null Hypothesis

		ϵ		
		1/3	1/4	1/5
Nominal Level of Test	.10	.40	.49	.55
	.05	.24	.31	.35
	.01	.07	.09	.11

Consider a test with the standard 5% Type I error rate. Searching between the 10th and 90th percentile would result in a realized Type I error rate of 49%, an inflation of 10 times the presumed level. Compressing the interval of search ameliorates the problem only slightly. Searching in the interquartile range results in an error rate of 31%, and even highly significant results ($p = 0.01$) would exhibit almost 10 times the assumed level of error.

2.4 Motivating Example: Dynamic Investment under Credit Constraints

To illustrate how the MED arises in economic contexts, we study how dynamic consumption and investment choices with constrained borrowing respond to wealth and credit shocks. A key step is to specify who is constrained, which is isomorphic to determining the threshold level of wealth beyond which individuals are effectively unconstrained.

Individuals solve the following utility maximization problem across periods t and $t + 1$

$$\begin{aligned}
V(s) &:= \max_{\substack{s \in \mathbb{R} \\ h \geq 0}} \log C_t + \beta \cdot \log C_{t+1} \\
\text{s.t. } & C_t + s = \tau(h) \cdot W_t + A_i \\
& C_{t+1} = \pi(h) \cdot W_{t+1} + (1 + r)s \\
& s \geq \bar{B}
\end{aligned} \tag{7}$$

where τ and π are the cost and return functions associated with schooling level h . The first order

condition for saving ensures consumption is smoothed over time according to the Euler equation

$u'(C_t)(-1) + \beta u'(C_{t+1})(1+r) = 0$. This implies optimal saving is

$$\frac{\beta(1+r) [\tau(h)W_t + A_i] - \pi(h)W_{t+1}}{(1+r)(1+\beta)} = s^* \quad (8)$$

The no-borrowing condition applied to optimal saving pins down a threshold level of wealth $\tilde{A}_i$ below which individual i faces binding credit constraints

$$\tilde{A}_i \geq \frac{\pi(h^*)W_{t+1} - \beta(1+r)\tau(h^*)W_t}{\beta(1+r)} \quad (9)$$

where h^* is the optimal value of human capital investment. In this model, $A_i \geq \tilde{A}_i$ implies i is unconstrained while $A_i < \tilde{A}_i$ implies they are constrained. To define unique cutoff values that apply for all observations, we propose an alternative set of “coarse” thresholds on either side of $\tilde{A}$ by substituting the maximum and minimum values of any random variable that varies across individuals. This leads to $\tilde{A}^{\text{Lower}} < \tilde{A} < \tilde{A}^{\text{Upper}}$ where individuals with $A_i < \tilde{A}^{\text{Lower}}$ are the most constrained (poorest) while $\tilde{A}_i > \tilde{A}^{\text{Upper}}$ are the least constrained (richest).

To characterize optimal investment decisions, first consider individuals who are not constrained. Assume there is fraction $\delta \in (0, 1)$ of unconstrained people with s^* satisfying the Euler equation above. The first-order condition for investment implies h^* is independent of wealth A_i and only responds to the wage premium. In contrast, constrained individuals always have lower levels of education than the unconstrained, so that their education decisions depending on their overall wealth levels. For the share $(1 - \delta) \in (0, 1)$ who are constrained, optimal borrowing $s^{**} = \bar{B}$ is right at the boundary, and implies large returns to schooling. Solving the resulting first-order condition implies optimal constrained schooling h^{**} levels for constrained individuals respond

not only wage premium effects, but also wealth. Only these people will respond to wealth shocks by increasing their schooling levels, and in the process closing pre-existing gaps in education between constrained and unconstrained people.

3 Minimum Effective Dose (MED)

The problems highlighted in the previous section comprise the core motivation of the empirical approach that we present here. To begin, we lay out what we believe to be a set of minimal assumptions to identify the ATT. To do so, we propose a split sample approach where one part of the dataset is used to identify a suitable control group, and the other then utilizes this group to estimate the ATT. This procedure will also have a correct Type I error level as the researcher is constrained in their choice of d_r .

3.1 Existence Assumption

All empirical settings suffer from the fundamental problem of causal inference, as we are unable to observe treated and untreated observations in the same time period for any unit. In a standard difference-in-differences setting, assumptions on the progression of counterfactual untreated outcomes for a treated group identify causal effects by comparing outcomes for treated and untreated groups before and after a policy change. In our setting, inference is further constrained as these comparisons are not possible because *all* units are *either* treated or untreated in each time period.

One way to restore the standard DD setting is to assume that untreated outcomes are observed for some units, even if all of them were treated. We operationalize this by assuming that a *Minimum Effective Dose* exists —intuitively, this restricts the individual treatment effect function $\mu_i(d_i)$ to equal zero for units receiving a dose below some threshold d_c . Formally:

A5 Minimum Effective Dose (MED) exists: $\exists d_c \in \mathcal{D}$ s.t. $\forall D_i < d_c, Y_{i\tau}(D_i, 1) = Y_{i\tau}(D_i, 0)$

Note that we do not assume units that are untreated exist. Rather, we assume that low dose units do not exhibit a treatment response so that their treated and untreated outcomes are identical at the dose they experienced. Under Assumption A5, potential outcomes in (2) can be simplified to

$$Y_{i,t} = (1 - P_t)Y_{i,t}(D_i, 0) + P_t\{Y_{i,t}(D_i, 1)T_i + Y_{i,t}(D_i, 0)(1 - T_i)\} \quad (10)$$

where $T_i := 1\{D_i \geq d_c\}$ denotes the (effectively) treated group. As this expression illustrates, the assumption of an MED allows us to recover a standard binary difference-in-differences setting. Under the standard parallel trends assumption, a difference-in-differences estimator will recover the average treatment effect for all (effectively) treated units above the MED:

Proposition 2. *Suppose that assumptions A1-A3, A4 (Parallel Trends), and A5 (Minimum Effective Dose) hold. Then the binned difference-in-differences estimator using the MED d_c recovers the ATT.*

In our estimation, this is the most straightforward justification of the binarization approach that 30% of the papers in our metastudy pursue. However, as we point out here, it relies on the assumption of a minimum effective dose, as well as a parallel trends assumptions. Further, the DD estimator studied above is generally infeasible, as it relies on knowledge of the MED d_c .

3.2 First Stage: Mallik et al. (2011) MED Estimator

To overcome this, we propose a consistent estimator of d_c by leveraging advancements in the threshold estimation literature. The standard approach involves assuming a homogenous, parametric dose response function and estimating treatment effects and the threshold d_c simultaneously. For example, we could specify the dose response function as $\mu_i(D_i; \beta) = \beta_1 + \beta_2(D_i - d_c)$ reflecting a linear “partial linear dose” approach that allows for a jump (β_1) in treatment effects

beyond the cutoff (d_c) as well as a constant multiple effect given their dose (β_2). Then, the true parameters $(\beta_0, \beta_1, \beta_2, d_c)$ will solve:

$$\underset{\beta, d}{\operatorname{argmin}} \sum_{i=1}^N [\Delta Y_{it} - \beta_0 - \mu_i(D_i; \beta_1, \beta_2) \cdot \mathbb{1}\{D_i \geq d_c\}]^2 \quad (11)$$

For every choice of d_c , $(\beta_0, \beta_1, \beta_2)$ can be estimated via least squares. Searching over all d_c and choosing the parameter set minimizing the expression in (11) will yield the solution. This method will work when the dose response function is homogeneous, specified correctly, and meets certain regularity conditions. We suspect that these restrictions will be unpalatable to practitioners and instead propose a nonparametric approach that relaxes many of these restrictions.

Using (10), we can write observed outcomes in each time period as

$$\begin{aligned} Y_{i,\tau-1} &= Y_{i,\tau-1}(D_i, 0) \\ Y_{i,\tau} &= Y_{i,\tau}(D_i, 0) + [Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0)]T_i \end{aligned}$$

Taking the first difference of observed outcomes yields:

$$\begin{aligned} \Delta Y_{i,\tau} &= \Delta Y_{i,\tau}(D_i, 0) + [Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0)]T_i \\ &= \Delta Y_{i,\tau}(D_i, 0) + \mu_i(D_i) \mathbb{1}(D_i \geq d_c) \end{aligned}$$

Finally, we can rewrite the right hand side using conditional expectations:

$$\begin{aligned} \Delta Y_{i,\tau} &= \mathbb{E}[\Delta Y_{i,\tau}(D_i, 0) | D_i] + \mathbb{E}[\mu_i(D_i) | D_i] \mathbb{1}(D_i \geq d_c) + \varepsilon_i \\ &= \mathbb{E}[\Delta Y_{i,\tau}(D_i, 0)] + \mathbb{E}[\mu_i(D_i) | D_i] \mathbb{1}(D_i \geq d_c) + \varepsilon_i \end{aligned}$$

Where the second line follows from policy relevance [A3.2] and parallel trends [A4]. We've written the first difference of observed outcomes as a standard threshold response model with an aggregate dose response function of the form $\mathbb{E}[\mu_i(D_i)|D_i]$. In addition, for estimation to become feasible, we need to impose a “no-crossing” property:

$$\text{A6 No crossing: } \mathbb{E}[\mu_i(D_i)|D_i] > 0 \quad \forall D_i > d_c$$

Note that this is far weaker than assuming monotonicity; we need only that the aggregate dose response does not change sign. Under assumptions A1-A6, we are able to estimate d_c without placing any parametric restrictions on $\mathbb{E}[\mu_i(D_i)|D_i]$ using the method in (Mallik et al., 2011), which we outline here. The author consider a general set-up of the form

$$Y_{ij} = \mu(X_i) + \epsilon_{ij}$$

Here, i indexes a set of dose values $\{X_i\}_{i=1,\dots,n}$ and $j = 1, \dots, J$ indexes each of the J observations at each dose. We consider a discrete setting, though this approach also works for a continuous dose support. The dose response function, $\mu(\cdot)$, is assumed to be constant at value τ for $d \leq d^0$, then strictly positive after d^0 .

Mallik et al. (2011) are able to estimate d^0 without estimating $\mu(\cdot)$ by leveraging a series of pairwise comparisons. Assume that we have access to an estimate $\hat{\tau}$ of τ , and let $\bar{Y}_i$ denote the average Y_{ij} at dose i . Consider testing the null hypothesis that these means are equivalent, $\bar{Y}_i = \hat{\tau}$, against the alternative $\bar{Y}_i > \hat{\tau}$. If $X_i \leq d^0$, the null hypothesis will be true, and if $X_i > d^0$, the null hypothesis will be false. Mallik et al. (2011) utilize approximate p-values of these hypothesis tests

$$P(D_i) = 1 - \Phi \left\{ \sqrt{m} \frac{\bar{Y}_i - \hat{\tau}}{\hat{\sigma}(X_i)} \right\}$$

Where $\hat{\sigma}(X_i)$ is an estimate of the standard error of Y_{ij} at dose X_i . Under the null, these p-values converge in distribution to Uniform[0, 1], and when it is not, they converge in probability to 0. As a result, d^0 will minimize the a “stump function” equal to 1/2 before d^0 and 0 after d^0 , given by:

$$\hat{d}^0 := \arg \max_{d^0 \in [0,1]} \sum_{i:D_i \leq d^0} \left[P(D_i) - \frac{1}{2} \right]^2 + \sum_{i:D_i > d^0} [P(D_i)]^2$$

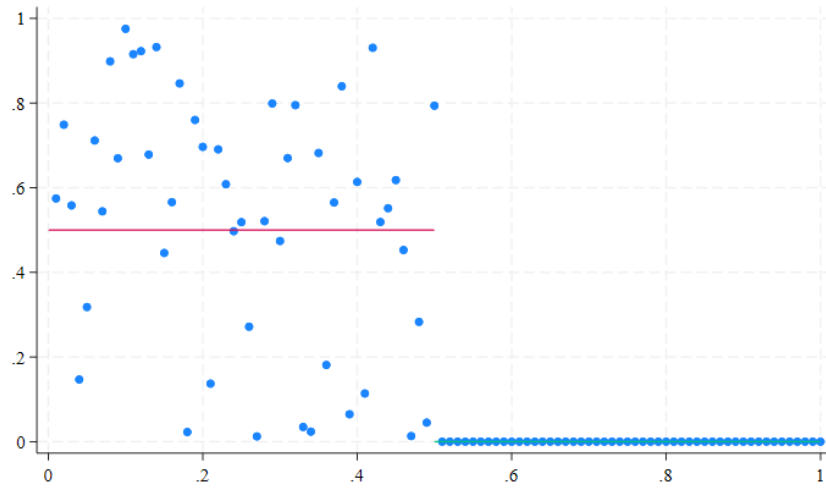
Mallik et al. (2011) show that this procedure consistently estimates $\hat{d}^0$. The rate of convergence is given by $n^{-1/(2k+1)}$, where k is the cusp of $\mu(\cdot)$ around the cutoff, as long as $\hat{\tau}$ converges at the parametric rate (Mallik et al., 2013). As a result, the smoother that $\mu(\cdot)$ is at d^0 , the slower this procedure converges; while the estimator is consistent, it exhibits a right bias in finite samples. Figure 1a plots the limiting distribution of p-values, while Figure 1b plots a common finite sample distribution, illustrating the source of right-bias.

We argue that this methodology can be used to estimate d_c . In our application, $\tau = \mathbb{E}[\Delta Y_{i,\tau}(D_i, 0)]$, and we propose using observations at the lowest dose d_l to estimate $\hat{\mathbb{E}}[\Delta Y_{i,\tau}(D_i, 0)]$. Under the assumption that an MED exists, which is necessary for inference, it follows that this dose must be untreated. Under assumptions A1-A6, all of the technical conditions are met, and we can obtain a consistent estimator of d_c , given by $\hat{d}_c$. However, we are ultimately concerned with estimation of the ATT, which we turn to next.

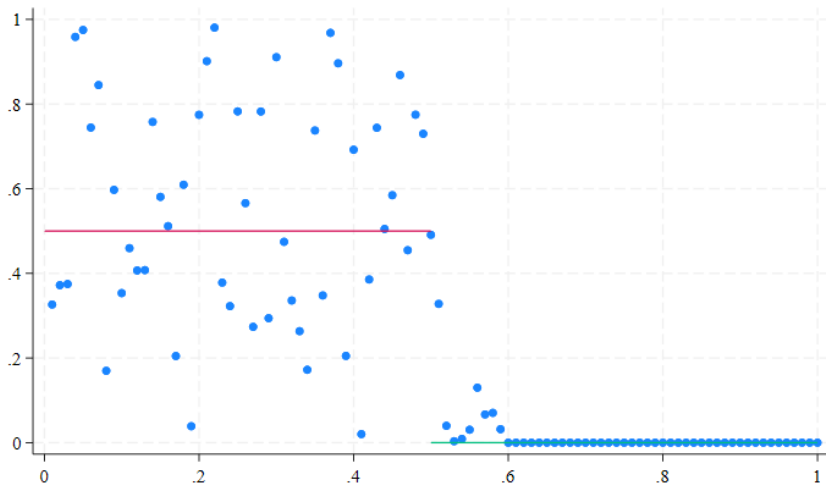
3.3 Second Stage: Least Squares Estimator of ATT

Given that we estimate the MED cutoff in a first stage using a random subset of the full data which does not overlap with observations in the second stage, we assume in this section that we have

Figure 1: Visualizing the p -value Stump Function



(a) Distribution in the Limit



(b) Distribution in Finite Samples

This figure plots a visualization of what the distribution of p -values should look like for a dose distribution over $[0, 1]$ where the cutoff is at 0.5. We plot a straight line at $1/2$ before the cutoff and at 0 after the cutoff to illustrate where the best fit line should lie.

access to a consistent estimator $\hat{d}_c$ of d_c . We then define the least squares binned estimator

$$\Delta Y_{i,t} = \beta_0 + \beta_1 \cdot 1\{D_i > \hat{d}_c\} + \Delta \epsilon_{i,t} \quad (12)$$

From Proposition 3, this two-step M-estimator identifies the ATT when $\hat{d}_c = d_c$

$$\sum_{i=1}^n (\Delta Y_{i,t} - \beta_0 - \beta_1 \cdot 1\{D_i > \hat{d}_c\})^2$$

And using standard arguments, we show that this estimator consistently estimates the ATT:

Proposition 3. *Suppose that assumptions A1-A3, A4 (Parallel Trends), and A5 (Minimum Effective Dose) hold. Then OLS estimation of (12) using the first-step estimator $\hat{d}_c$ is consistent for the ATT.*

3.4 Standard Error Estimation

Constructing asymptotically valid standard errors is a challenging task. It is well-known that conventional OLS standard errors from the second-step are not asymptotically valid, as they don't reflect uncertainty in the value of the nuisance parameter (Newey and McFadden, 1994). One approach, used in Hansen (2000), uses a Bonferroni correction to construct conservative but asymptotically valid confidence intervals. Threshold estimators generally have non-standard asymptotic distributions that cannot be estimated using a standard non-parametric bootstrap (Seijo and Sen, 2011). Mallik et al. (2013) derive asymptotically valid confidence intervals for the approach above, though they contain nuisance parameters that are difficult to estimate. Given this confidence interval, the union of standard OLS confidence intervals as a function of all values in the $\hat{d}_c$ confidence interval will have proper coverage, but will also likely overstate uncertainty in the second stage.

Alternatively, we can use the structure of the second stage estimator to tighten these standard

errors. Following [Randles \(1982\)](#), under certain conditions, we can use the Taylor expansion

$$\sqrt{n}(\hat{\beta}_1(\hat{d}(c)) - \beta_1(d_c)) = \sqrt{n}(\hat{\beta}_1(d(c)) - \beta_1(d_c)) + \sqrt{n}(\hat{d}_c - d_c)\beta'_1(d_c) + o_p(1)$$

which converges to a normal distribution under standard OLS assumptions. From work in [Chan \(1993\)](#), [Hansen \(1999\)](#), and [Hansen \(2000\)](#), it is well known that in a parametric threshold model with a discontinuity at the threshold, error from estimation of d_c is not of first order importance in the asymptotic distribution of $\hat{\beta}_1$ because $\hat{d}_c$ converges at a n^{-1} rate, faster than the parametric convergence rate of the slope coefficients of $n^{-1/2}$. As a result, inference in the second stage can proceed as if d_c is known and confidence interval coverage will still be asymptotically valid.⁷ Thus, under a jump discontinuity at d_c , inference in the second stage can be done using either usual OLS standard errors or the non-parametric bootstrap, ignoring any uncertainty in the first stage. We show with simulations that the bootstrap achieves proper coverage of the ATT.

This approach generally depends on the cusp of the DRF at the cutoff. If the DRF is continuous at d_c , $\hat{d}_c$ will converge at a rate below the parametric rate and will affect the limiting distribution of $\hat{\beta}_1$. In these cases we propose using uneven sample sizes for the first and second stage to recover favorable convergence rates. Since estimating the MED is much harder than estimating the ATT, we suggest using more data for the former relative to the later. Assuming the first derivative of the DRF is discontinuous at d_c , [Zhou et al. \(2010\)](#) define an estimator which converges at a rate of $n^{-1/3}$. Denote the MED sample n_{MED} and the ATT sample n_{ATT} where $n_{MED} + n_{ATT} = n$. If we choose sample sizes such that $n_{MED} = n_{ATT}^{3/2}$, $\hat{d}_c$ will converge at an $n_{ATT}^{-1/2}$ rate. In this case, $\sqrt{n_{ATT}}(\hat{\beta}_1(\hat{d}(c)) - \beta_1(d_c))$ will have a normal distribution in the limit, which can be approximated with the standard non-parametric bootstrap.

⁷Surprisingly, the nonparametric wavelet estimator in [Zhou et al. \(2010\)](#) also achieves this convergence rate.

3.5 Simulation Evidence

We first compare the split-sample estimator to dichotomizing at some researcher-specified percentile, in this case $d_r = 0.7$ which does not equal the true minimum effective dose (MED) parameter $d_c = 0.5$ at the median. As a benchmark, we also compute the Oracle estimator which compares units above and below the true MED value in the whole sample. The sample consists of $i = \{1, \dots, N\}$ units observed across $j = \{1, \dots, M\}$ dose values, and we assume a constant treatment effect for all units with dose values above the median. The following algorithm constructs the simulations in Figures 2 and 3, which plot the distribution of the split-sample estimator.

1. Set $d_L = 0$, $d_c = 0.5$, $d_r = 0.7$, and $d_U = 1$
2. Set $\beta_{\text{Jump}} = 1$ in DRF $\mu(D_i) := 1\{D_i > d_c\} \cdot \beta_{\text{Jump}}$
3. Draw $\vec{\Delta\varepsilon}_s \sim F_\varepsilon(\mathbb{R})^N$ and $\vec{D}_s \sim F_D[d_L, d_U]^M$ where $F_\varepsilon = N(0, 1)$ and $F_D = \text{UNIF}[0, 1]$
4. Compute $\Delta Y_{i,s} := \mu(D_i) + \Delta\varepsilon_{i,s}$
- 5.1. Compute $\hat{\beta}_{0,s}^{\text{Binarize}} := \hat{E}[\Delta Y_{i,s} \mid D_i \geq d_r] - \hat{E}[\Delta Y_{i,s} \mid D_{i,s} \leq d_r]$
- 5.2. Compute $\hat{\beta}_{0,s}^{\text{Oracle}} := \hat{E}[\Delta Y_{i,s} \mid D_i \geq d_c] - \hat{E}[\Delta Y_{i,s} \mid D_{i,s} \leq d_c]$
- 6.1. Split sample randomly into two groups
- 6.2. In first group, implement P-value estimator to compute $\hat{d}_{c,s}$
- 6.3. In second group, compute $\hat{\beta}_{0,s}^{\text{Split}}(\hat{d}_{c,s}) := \hat{E}[\Delta Y_{i,s} \mid D_i \geq \hat{d}_{c,s}] - \hat{E}[\Delta Y_{i,s} \mid D_{i,s} \leq \hat{d}_{c,s}]$
7. Repeat 1-6 for simulations $s \in \{1, \dots, S\}$
8. Plot distributions of $\{\hat{\beta}_{0,s}^{\text{Binarize}}, \hat{\beta}_{0,s}^{\text{Oracle}}, \hat{d}_{c,s}, \hat{\beta}_{0,s}^{\text{Split}}(\hat{d}_{c,s})\}$

We see that both the first stage estimator of the MED and the second stage estimator of the ATT are consistent. Splitting the sample to determine the optimal cutoff offers an improvement over researcher dichotomization at some percentile, albeit at the cost of some precision, as the figures confirm that a smaller second-stage sample increases variance in estimation.

Figure 2: Distribution of Split Sample Estimator of ATT and MED

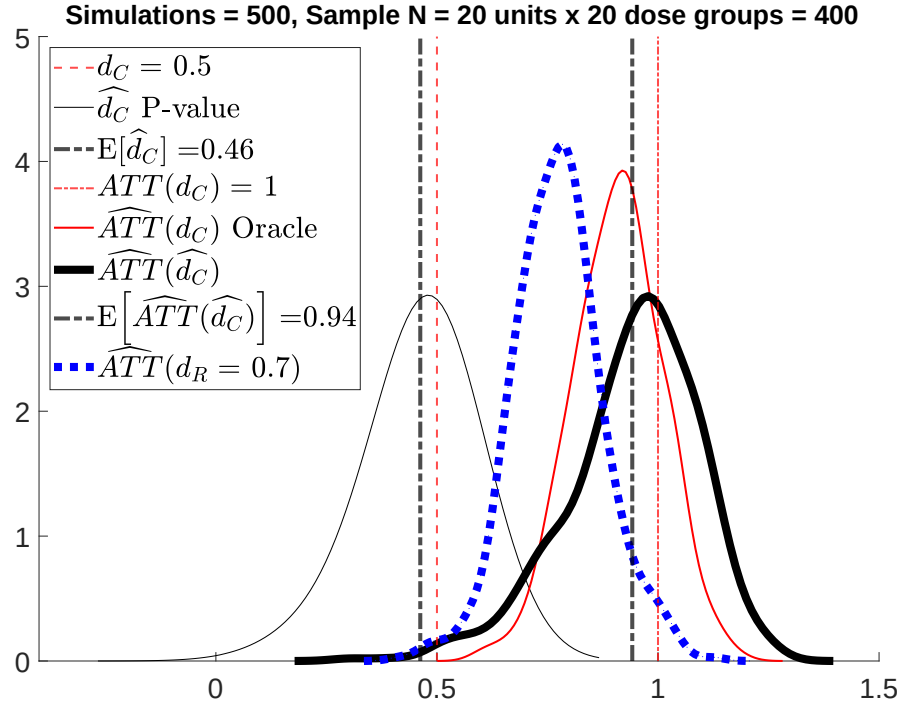
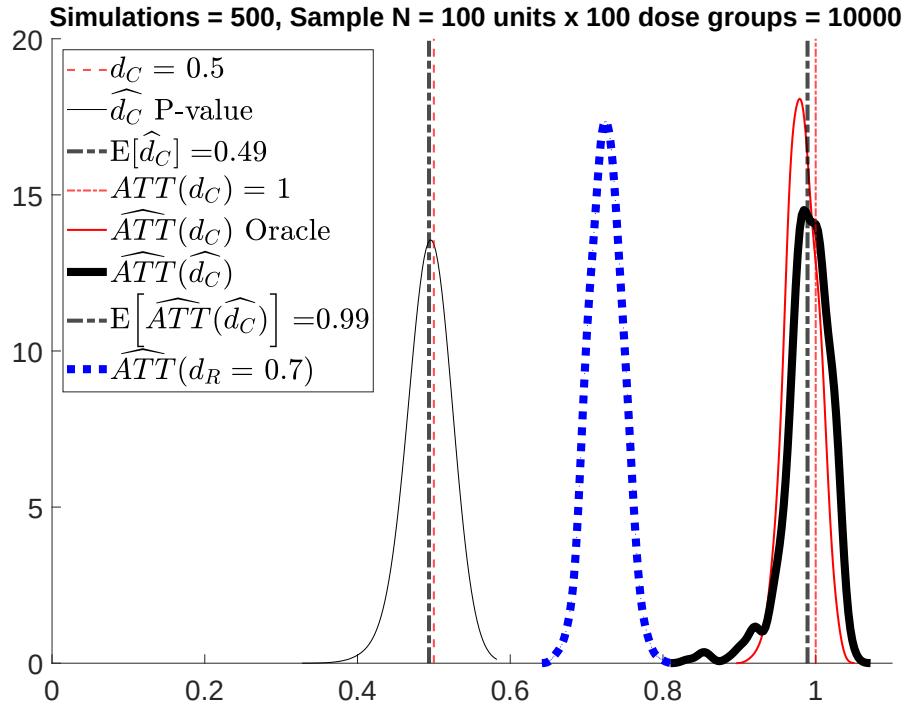


Figure 3: Distribution of Split Sample Estimator of ATT and MED



The following algorithm describes our simulation of the bootstrap estimator of the standard error under parametric assumptions on the dose response function: constant, linear, and quadratic.

1. Set size $\alpha = 0.05$, $d_L = 0$, $d_c = 0.5$, $d_r = 0.7$, and $d_U = 1$
2. Set $\beta_{\text{Jump}}, \beta_{\text{Lin}}, \beta_{\text{Quad}}$ in DRF $\mu(D_i) := 1\{D_i > d_c\} \cdot [\beta_{\text{Jump}} + \beta_{\text{Lin}}(D_i - d_c) + \beta_{\text{Quad}}(D_i - d_c)^2]$
3. Draw $\vec{\Delta\varepsilon}_s \sim F_\varepsilon(\mathbb{R})^N$ and $\vec{D}_s \sim F_D[d_L, d_U]^M$ where $F_\varepsilon = N(0, 1)$ and $F_D = \text{UNIF}[0, 1]$
4. Compute $\Delta Y_{i,s} := \mu(D_i) + \Delta\varepsilon_{i,s}$
- 5.1. Split sample randomly into two groups. Within each, draw a random set of bootstrap subsamples $b = \{1, \dots, B\}$ with replacement and equal in length to the original dataset
- 5.2. In first group, implement P-value estimator $\hat{d}_{c,s}(b)$ across all bootstrap samples
- 5.3. In second group, compute $\hat{\beta}_{0,s}^{\text{Split}}(\hat{d}_{c,s}(b)) := \hat{E} [\Delta Y_{i,s} \mid D_i \geq \hat{d}_{c,s}(b)] - \hat{E} [\Delta Y_{i,s} \mid D_{i,s} \leq \hat{d}_{c,s}(b)]$
- 6.1. Compute the 2.5 and 97.5 percentiles of the bootstrap distribution $\hat{\beta}_{0,s}^{\text{Split}}(\hat{d}_{c,s}(b))$
- 6.2. Compute $I(s) := 1 \left\{ ATT \in (p_\beta^{2.5}, p_\beta^{97.5}) \right\}$ which indicates whether the true value of the $ATT := E[\mu(D_i)]$ is contained in the bootstrap confidence interval
7. Repeat 1-6 for simulations $s \in \{1, \dots, S\}$
8. Compute average of $I(s)$

Figure 4 below plots simulated coverage of the bootstrap estimator of the standard error, across various parameter and sample size values. It shows that the first stage MED estimator converges fast enough to not markedly influence variance in the second stage, as most confidence intervals cover the true ATT value about 95% of the time given size $\alpha = 0.05$. Moreover, this result holds for dose-response functions that are smooth at the cutoff, in addition to those with a jump discontinuity, for which we have theoretical basis to expect proper coverage.

Figure 4: Simulated Coverage of Split Sample Estimator of ATT

Dose Groups	Units	$(\beta_{\text{Jump}}, \beta_{\text{Lin}}, \beta_{\text{Quad}}) =$				
		(1,0,0)	(1,1,0)	(1,0,1)	(0,1,0)	(0,0,1)
5	5	0.96	0.96	0.96	0.97	0.96
	10	0.95	0.96	0.95	0.97	0.95
	20	0.97	0.96	0.97	0.98	0.97
10	5	0.97	0.96	0.97	1.00	0.99
	10	0.96	0.95	0.95	0.99	0.98
	20	0.94	0.93	0.94	0.99	0.97
20	5	0.97	0.96	0.96	1.00	1.00
	10	0.94	0.93	0.94	0.98	0.95
	20	0.94	0.93	0.93	0.96	0.96

Results computed across 500 simulations, and 500 bootstrap replications of the data-generating process.

4 Extensions

4.1 Testing the MED Existence Assumption

One concern with estimating d_c is that the estimation procedure will always yield an estimate, even if there does not exist a minimum effective dose. To address this, we propose the addition of a statistical test that all units below $\hat{d}_c$ are, in fact, untreated. Since this variation is unused in the sample estimating the ATT, this should be the best place to test this hypothesis.

Consider the following quasi-randomization test (using the terminology provided by [Zhang and Zhao \(2023\)](#)). Under the null hypothesis that $\hat{d}_c < d_c$, the data generating process is invariant to arbitrary dose permutations for units below $\hat{d}_c$. It is easy to verify that the randomization

hypothesis holds in this environment (Lehmann and Romano, 2022):

Proposition 4. *Suppose that assumptions A1-A5 hold. Consider the subset of the data $\{Y_{i,\tau}, Y_{i,\tau-1}, D_i\}_{i=1}^N$ where $D_i < \hat{d}_c$. Under the assumption that $\hat{d}_c < d_c$ the data are invariant under arbitrary permutations of assigned doses $\{D_i\}_{i=1}^N$*

By the randomization hypothesis, this will have proper Type-I coverage with any test statistic, so we focus on choosing one with the desirable power properties. A natural choice is a simple linear regression on the dose. Corollary 3.1 in Callaway et al. (2024) shows that this estimator recovers

$$\int_{d_L}^{d_U} w(\ell) \cdot \left. \frac{\partial E[\Delta Y_{it} | D_i = d]}{\partial d} \right|_{d=\ell} d\ell$$

The weights $w(\ell)$ are non-negative and integrate to one. This design has power against monotonic alternatives, but fails if the derivative of the conditional expectation $E[\Delta Y_{it} | D_i = d]$ changes signs.⁸ Putting this together, we propose the following procedure:

1. Split dataset randomly into two parts
 - (a) $\mathcal{D}_{1,n}$ to estimate d_c
 - (b) $\mathcal{D}_{2,n}$ to estimate ATT
2. Using $\mathcal{D}_{1,n}$, produce estimate $\hat{d}_c$
 - (a) Set $\hat{\tau} = \Delta \bar{Y}_{i\tau} |_{D_i = d_i}$
 - (b) Calculate approximate p -values at all other dose values
 - (c) Choose $\hat{d}_c$ that minimizes stump function
3. Using $\mathcal{D}_{2,n}$, test assumption that MED exists
4. Using $\mathcal{D}_{2,n}$, conduct inference on ATT
 - (a) Estimate ATT using binned design

⁸Future work will explore the use of copula-based test statistics for independence (Genest and Rémillard, 2004).

4.2 Dichotomizing at Researcher-Specified Percentile

The results in this section are based on a misspecification analysis where the hope is that, even without knowledge of the true MED d_c , throwing away dose variation can nevertheless recover an attenuated version of the ATT. However, we show that this is generally not true, and only holds under restrictive assumptions on the heterogeneity of the dose response function across individuals. To this end, we introduce a formal assumption of homogeneity:

A7 Dose Response Function Homogeneity: Dose response functions are the same across units and given by $\mu_i(D_i) = \mu(D_i) \forall i$.

In the simplest case, the dose response function is identical for all units, and the dose response function is constant, so that $\mu(D_i) = \beta$. In this case, even without knowledge of d_c , a binned design will recover an attenuated version of the ATT:

Proposition 5. *Suppose that assumptions A1-A5 hold, and that the DRF is homogeneous (A6 holds) where $\mu(D_i) = \beta$. Then, the binned estimator under an arbitrary cutoff point d_r will give:*

$$\hat{b}_1^{BIN} \rightarrow^p ATT \times \min \left\{ \frac{1 - F(d_c)}{1 - F(d_r)}, \frac{F(d_c)}{F(d_r)} \right\}$$

We find that the researcher guess leads to an attenuated estimate of the treatment effect on the treated (β), akin to the result of classical measurement error. If the researcher guess is too low ($d_r < d_c$), the intuition is very straightforward. Some of the “treatment” units are actually untreated - by definition, they receive a treatment effect of 0 - and as a result, the estimated treatment effect is a mixture of β and 0. If the researcher guess is too high ($d_c > d_r$), the result is more convoluted. Some of the “control” units are actually treated, and this contamination works to attenuate the estimated treatment effect. In addition, “treatment” units are a higher dose subset (above d_r) of

the true treated group (above d_c); since $\mu_i(D_i)$ does not depend on the dose (or individual), this conditioning does not impact the average difference in outcomes in the “treated” group, and we obtain the symmetric result in Proposition 4.

We can also replicate this result if dose assignment is randomized:

A8 Random Dose Assignment: Dose assignment is independent of the observed dose response

so that $D_i \perp \mu_i(D_i)$

Proposition 6. *Suppose that assumptions A1-A5 hold, and that the dose is assigned randomly (A8 holds).*

Then, the binned estimator under an arbitrary cutoff point d_r will give:

$$\hat{b}_1^{BIN} \rightarrow^p ATT \times \min \left\{ \frac{1 - F(d_c)}{1 - F(d_r)}, \frac{F(d_c)}{F(d_r)} \right\}$$

4.2.1 Example: (ATT)enuation Bias at Worst?

The attenuation result breaks down if we begin to relax any part of the assumption of a uniform binary dose response function, in which case the average difference in outcomes above d_r is not identical to that for units above d_c . To see this, consider an example with a binary dose response function for all units, where the level of the response might vary across units; that is, $\mu_i(D_i) = \beta_i$. For simplicity, suppose the dose distribution is $U[0, 1]$, the treatment cutoff $d_c = 1/2$ for all units, and dose response functions are given by:

$$\mu_i(D_i) = \begin{cases} 0 & \text{if } D_i < 1/2 \\ 6 \times \mathbb{1}(D_i \geq 1/2) & \text{if } D_i \in [1/2, 3/4] \\ 24 \times \mathbb{1}(D_i \geq 1/2) & \text{if } D_i \in [3/4, 1] \end{cases} \quad (13)$$

In this example, all units above $d_c = 1/2$ are treated, and the average treatment effect on the treated is 15, which will be estimated properly at the correct cutoff $d_c = 1/2$. Now, we can consider alternate researcher cut-off choices. Table 2 shows the estimated treatment effect on the treated, $\widehat{ATT}$, as a function of the researcher choice of threshold. For a guess that is too low (e.g. $1/4$), we will always recovered an attenuated treatment effect, as the researcher treatment group is mixing all treated units with some fraction of the control units. However, for a guess that is too high (e.g. $3/4$), we are biased upwards, as higher dose units have a larger (flat) dose response. This

Table 2: Binned Estimator Under Selection

d_r	$[0, 1/4)$	$[1/4, 1/2)$	$[1/2, 3/4)$	$[3/4, 1)$	$\widehat{ATT}$
$1/4$	Control	Treatment	Treatment	Treatment	10
$1/2$	Control	Control	Treatment	Treatment	15
$3/4$	Control	Control	Control	Treatment	22

bias occurs because, unlike in Proposition 4, the average change in outcomes above $3/4$ is higher than the average change in outcomes above $1/2$. In our simple example, this is because the dose response β_i is positively correlated with the dose D_i .

If homogeneity fails and we are not willing to assume that dose assignment is random, the above example illustrates that even with a very simple dose response function our estimator will have an unknown bias. Unfortunately, this issue cannot be dealt with even with the assumption of a homogenous dose response function. To see this, consider the following functional form:

$$\mu(D_i) = \begin{cases} 0 & \text{if } D_i < 1/2 \\ 6 & \text{if } D_i \in [1/2, 3/4) \\ 24 & \text{if } D_i \in [3/4, 1) \end{cases} \quad (14)$$

This gives rise to a set of observed outcomes identical to that in (13), and as a result, the ATT

estimates in Table 2 will remain the same. While there may be a set of dose response functions that only admit attenuation bias, it is troubling that an example that only takes on two values can lead to bias of an unknown sign.

4.3 What if MED is Unknown

We now consider what, if anything, is identified without exact knowledge of the true cutoff under the assumption of a minimum effective dose. Clearly, the ATT is a poor estimand for this type estimator, as the practitioner does not know whether or not d_r is close to d_c . One way to solve this issue is to instead target the *binned average treatment effect on the treated* (BATT) which averages individual treatment effects over a range of doses.

The BATT estimand frames our discussion of identification in this case. While this terminology is new, the concept is not; BATTs extend the notion of a dose-specific average treatment effect on the treated (ATT-d) utilized in Callaway et al. (2024). This estimand is useful as the treatment group matches that explicitly defined in a binned estimator used in practice. We recommend that practitioners target a BATT instead of an ATT, which has two key advantages.

First, since a BATT is defined by the binned treatment groups specified by the researcher, the treated group ($D_i \geq d_r$) is always identified correctly by the researcher in the data. The only risk, then, is that control units ($D_i < d_r$) are not below the MED and might lead to a biased estimator.

The second advantage of targeting a BATT is a corollary of the first: if the researcher specified threshold is below the MED ($d_r < d_c$), then the binned estimator will be unbiased for $\text{BATT}[d_r, \infty]$ with no further assumptions, which is established in the next proposition.

Proposition 7. *Suppose that assumptions A1-A5 hold. Consider the binned estimator under an arbitrary cutoff point d_r . If $d_r < d_c$, this estimator will be unbiased for $\text{BATT}[d_r, \infty]$.*

Under the assumption of a minimum effective dose, $BATT[d_c, \infty]$ will be equivalent to the average treatment effect on the treated (ATT), and we center our discussion of inference under this assumption below on targeting this causal estimand.

4.3.1 Monotonicity Assumption

Targetting the $BATT[d_r, \infty]$ estimand has several advantages. First, in many settings, this estimand is of direct policy interest; for example, we could estimate average treatment effects for families earning above the poverty line.⁹ Second, by construction, the treatment group is defined correctly by the binned estimator using d_r as a cutoff; in contrast, using an identical binned estimator to estimate the ATT would involve treatment units being intentionally misclassified as controls. Finally, under a natural assumption of treatment effect monotonicity, the binned estimator will always recover an attenuated version of $BATT[d_r, \infty]$.

The following monotonicity assumption is natural in any setting with a continuous treatment. If we do not expect individual treatment effects to be weakly increasing in the dose, then high-low comparisons would be a bizarre route to pursue, in either a binned or continuous setting.

A6 Monotonicity: For any pair of dose values such that $d < d'$, $\mu(d|d) \leq \mu(d'|d')$

We assume here that the dose-specific ATT is weakly increasing in the dose. Notice that this might be true if individual dose response curves are increasing in the dose, or if units receiving higher doses have a stronger dose response, or perhaps some combination of both.

A standard difference-in-differences estimator will utilize the low dose “control” group to difference out the change in the average untreated outcome, but will also subtract whatever treatment effect the low dose group experiences. As long as this term is the same sign but smaller in magni-

⁹Note that setting d_r to an empirical percentile generally does not achieve this, a point we return to later.

tude than the treatment effect of interest, we will recover an attenuated version of $BATT[d_r, \infty]$:

Proposition 8. *Suppose that assumptions A1-A3, A4.2 (Parallel Trends), and A6 (Monotonicity) hold.*

Then the binned difference-in-differences estimator recovers an attenuated version of the target BATT.

Our contribution here is to clarify a (potentially restrictive) assumption that is necessary for this result to hold. To this end, the proof of Proposition 8 shows that for any given d_r , the weaker assumption of monotonicity of the “treatment” and “control” BATTs is also sufficient. However, since the choice of d_r is arbitrary, we choose our assumption of monotonicity such that it accommodates any comparison of high dose and low dose groups. We verify this result formally:

Proposition 9. *The following assumptions are equivalent:*

- (i) $\forall d, d' \in \mathcal{D} \text{ s.t. } d < d', \mu(d|d) \leq \mu(d'|d')$
- (ii) $\forall D, D' \subset \mathcal{D} \text{ s.t. } \sup\{D\} \leq \inf\{D'\}, BATT[D] \leq BATT[D']$

4.3.2 Full Dose Regression

70% of papers estimate a full dose regression, which is equivalent to the bivariate regression:

$$\Delta Y_{i,t} = \alpha + \beta \cdot D_i + \Delta \epsilon_{i,t} \quad (15)$$

As [Callaway et al. \(2024\)](#) point out, interpreting $\hat{\beta}$ in this model is difficult and varies across practitioners. Of particular interest for our results is equation (3.1) which states

$$\hat{\beta} \xrightarrow{p} \frac{\mathbb{E}[w_1^{bin}(D_i)\Delta Y_{it}|D_i > \mathbb{E}[D_i]] - \mathbb{E}[w_0^{bin}(D_i)\Delta Y_{it}|D_i < \mathbb{E}[D_i]]}{\mathbb{E}[w_1^{bin}(D_i)D_i|D_i > \mathbb{E}[D_i]] - \mathbb{E}[w_0^{bin}(D_i)D_i|D_i < \mathbb{E}[D_i]]}$$

where the weights are given by

$$w_1^{bin}(d) = \frac{|d - \mathbb{E}[D_i]|}{\mathbb{E}\left[|D_i - \mathbb{E}[D_i]| \middle| D_i > \mathbb{E}[D_i]\right]}$$

$$w_0^{bin}(d) = \frac{|d - \mathbb{E}[D_i]|}{\mathbb{E}\left[|D_i - \mathbb{E}[D_i]| \middle| D_i \leq \mathbb{E}[D_i]\right]}$$

The full dose estimator recovers a Wald-like parameter, taking the difference between a weighted average of the outcome above and below the mean, and scaling by the average distance in doses between these two groups. The key issue for interpretation is that $\hat{\beta}$ is not invariant to scaling of the dose variable, through its impact on the denominator. In the appendix, we show it is always possible to apply a transformation to the dose distribution so the denominator equals one:

$$\hat{\beta}^{scaled} \rightarrow_p \mathbb{E}[w_1^{bin}(D_i)\Delta Y_{it}|D_i > \mathbb{E}[D_i]] - \mathbb{E}[w_0^{bin}(D_i)\Delta Y_{it}|D_i < \mathbb{E}[D_i]]$$

The full dose regression in (15) is thus similar to a binned design where the cutoff is chosen to be the mean of the dose distribution. Scaling does not change “treatment” and “control” groups, nor does it affect the weights. See appendix for details. Up to a linear transformation of the dose, these estimators only vary in their choice of weights.

5 Conclusion

In the paper, we construct a potential outcomes framework for empirical settings where all units are treated. We formalize baseline assumptions needed for inference - the existence of a minimum effective dose, which implies a subset of units experience untreated outcomes. Leveraging this assumption, we propose a non-parametric estimator of the ATT that first estimates the MED in a

hold-out sample, and we provide auxiliary statistical tests of the assumption that the MED exists.

References

- Altman, D. G., B. Lausen, W. Sauerbrei, and M. Schumacher (1994). Dangers of using “optimal” cutpoints in the evaluation of prognostic factors. *Journal of the National Cancer Institute* 86(11), 829–835.
- Angrist, J. D. and J.-S. Pischke (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton, NJ: Princeton University Press.
- Bound, J., C. Brown, and N. Mathiowetz (2001). Measurement error in survey data. *Handbook of Econometrics* 5, 3705–3843.
- Brenner, H. (1997). A potential pitfall in control of covariates in epidemiologic studies. *Epidemiology* 9(1), 68–71.
- Brodeur, A., N. Cook, and A. Heyes (2020, November). Methods matter: p-hacking and publication bias in causal analysis in economics. *American Economic Review* 110(11), 3634–60.
- Butts, K. (2022). Difference-in-differences with geocoded microdata. *Journal of Urban Economics*.
- Callaway, B., A. Goodman-Bacon, and P. H. C. Sant’Anna (2024). Difference-in-differences with a continuous treatment. Working Paper.
- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*.
- Card, D. (1992). Using regional variation in wages to measure the effects of the federal minimum wage. *Industrial and Labor Relations Review*, 22–37.
- Chan, K. S. (1993). Consistency and limiting distribution of the least squares estimator of a threshold autoregressive model. *The Annals of Statistics* 21(1), 520–533.

- de Chaisemartin, C. and X. D'Haultfoeuille (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *Journal of Econometrics*.
- de Chaisemartin, C., X. D'Haultfoeuille, F. Pasquier, and G. Vazquez-Bare (2023). Difference-in-differences estimators for treatments continuously distributed at every period. *Working Paper*.
- Genest, C. and B. R  millard (2004). Test of independence and randomness based on the empirical copula process. *Test* 13, 335–369.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*.
- Hansen, B. E. (1996). Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica* 64(2), 413–430.
- Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing, and inference. *Journal of Econometrics* 93, 345–368.
- Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica* 68(3), 575–603.
- Hansen, B. E. (2017, April). Regression Kink With an Unknown Threshold. *Journal of Business & Economic Statistics* 35(2), 228–240.
- Imbens, G. W. and D. B. Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. New York, NY: Cambridge University Press.
- Lausen, B. and M. Schumacher (1992). Maximally selected rank statistics. *Biometrics* 48(1), 73–85.
- Lehmann, E. L. and J. P. Romano (2022). *Testing Statistical Hypotheses*. Cham, Switzerland: Springer.
- MacCallum, R. C., S. Zhang, K. J. Preacher, and D. D. Rucker (2002). On the practice of dichotomization of quantitative variables. *Psychological Methods* 7(1), 19–40.

- Mallik, A., M. Banerjee, and B. Sen (2013). Asymptotics for p-value based threshold estimation in regression settings. *Electronic Journal of Statistics* 7, 2477–2515.
- Mallik, A., B. Sen, M. Banerjee, and G. Michailidis (2011). Threshold estimation based on a p-value framework in dose-response and regression settings. *Biometrika* 98(4), 887–900.
- Miao, K., K. Li, and L. Su (2020). Panel threshold models with interactive fixed effects. *Journal of Econometrics* 219, 137–170.
- Miao, K., L. Su, and W. Wang (2020). Panel threshold models with latent group structures. *Journal of Econometrics* 214, 451–481.
- Miller, R. and D. Siegmund (1982). Maximally selected chi square statistics. *Biometrics* 38(4), 1011–1016.
- Newey, W. K. and D. McFadden (1994). Large sample estimation and hypothesis testing. In R. F. Engle and D. McFadden (Eds.), *Handbook of Econometrics*, Volume 4 of *Handbook of Econometrics*, Chapter 36, pp. 2111–2245. Elsevier.
- Rambachan, A. and N. Shephard (2019). A nonparametric dynamic causal model for macroeconomics. Working Paper.
- Randles, R. H. (1982). On the asymptotic normality of statistics with estimated parameters. *The Annals of Statistics* 10(2), 462–474.
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*.
- Roth, J. and P. H. Sant’Anna (2021). When is parallel trends sensitive to functional form? *Working Paper*.

- Ruberg, S. J. (1989). Contrasts for identifying the minimum effective dose. *Journal of the American Statistical Association* 84(407), 816–822.
- Seijo, E. and B. Sen (2011). Change-point in stochastic design regression and the bootstrap. *Journal of the American Statistical Association* 39(3), 1580–1607.
- Seo, M. H. and Y. Shin (2016). Dynamic panels with threshold effect and endogeneity. *Journal of Econometrics* 195, 169–186.
- Sun, L. and S. Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*.
- Sun, L. and J. M. Shapiro (2022). A linear panel model with heterogeneous coefficients and variation in exposure. *Journal of Economic Perspectives*.
- Wang, Q. (2015). Fixed-effect panel threshold model using stata. *The Stata Journal* 15(1), 121–134.
- Wooldridge, J. (2021). Two-way fixed effects, the two-way mundlak regression, and difference-in-differences estimators. *Working Paper*.
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*. Cambridge, MA: MIT Press.
- Yitzhaki, S. (1996). On using linear regressions in welfare economics. *Journal of Business Economic Statistics* 14(4), 478–486.
- Zhang, Y. and Q. Zhao (2023). What is a randomization test? *Journal of the American Statistical Association* 118(544), 2928–2942.
- Zhou, Y., A. T. Wan, S. Xie, and X. Wang (2010). Wavelet analysis of change-points in a non-parametric regression with heteroscedastic variance. *Journal of Econometrics* 159(1), 183–201.

A Proofs

A.1 Proposition 1

Proof. Consider the probability limit of $\hat{E}[\Delta Y_{i\tau}|T_i = 1] - \hat{E}[\Delta Y_{i\tau}|T_i = 0]$, where $T_i = \mathbb{1}(d_i \geq d_r)$.

$$\begin{aligned}
\hat{E}[\Delta Y_{i\tau}|T_i = 1] - \hat{E}[\Delta Y_{i\tau}|T_i = 0] &= E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 0] \\
&= (E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 1]) \\
&\quad - (E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 0] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 1]) \\
&= (E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 1]) \\
&\quad - (E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 0] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 0]) \\
&= (E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) - |T_i = 1]) - (E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 0]) \\
&= \text{BATT}[d_r, d_u] - \text{BATT}[d_l, d_r]
\end{aligned}$$

□

A.2 Proposition 2

Proof. Consider the probability limit of $\hat{E}[\Delta Y_{i\tau}|T_i = 1] - \hat{E}[\Delta Y_{i\tau}|T_i = 0]$, where $T_i = \mathbb{1}(d_i \geq d_c)$.

$$\begin{aligned}
\hat{E}[\Delta Y_{i\tau}|T_i = 1] - \hat{E}[\Delta Y_{i\tau}|T_i = 0] &\rightarrow_p E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 0] \\
&= E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 0] \\
&= E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] - E[Y_{i,\tau}(D_i, 0) - Y_{i,\tau-1}(D_i, 0)|T_i = 1] \\
&= E[Y_{i,\tau}(D_i, 1)|D_i \geq d_c] - E[Y_{i,\tau}(D_i, 0)|D_i \geq d_c] = \text{ATT}
\end{aligned}$$

The second equality follows from the assumption of a minimum effective dose, and the third equality follows from the parallel trends assumptions. $\square$

A.3 Proposition 3

Proof. We show consistency of the following two-step M-estimator, where $\hat{d}_c$ is consistent for d_c :

$$\operatorname{argmin} \sum_{i=1}^n (\Delta Y_{i,t} - \beta_0 - \beta_1 \cdot 1\{D_i > \hat{d}_c\})^2$$

Following [Hansen \(2017\)](#), our target OLS parameters don't comprise a structural relationship, but rather minimize squared loss of an objective function of interest evaluated at the true d_c , so that

$$(\beta_0, \beta_1) = \operatorname{argmin}_{\hat{\beta}_0, \hat{\beta}_1} \mathbb{E}[(\Delta Y_{i,t} - \hat{\beta}_0 - \hat{\beta}_1 \cdot 1\{D_i > d_c\})^2]$$

As a result, the identification condition in [Wooldridge \(2010, p. 354\)](#) is met by assumption.

Assume that the OLS parameters (β_0, β_1) lie in some bounded, compact space Θ . By Assumption A2, then, the parameter space $\Theta \times \mathcal{D}$ is compact. Consider the function $a(Y_{i,t}, D_i, \beta_0, \beta_1, d_c) = (\Delta Y_{i,t} - \beta_0 - \beta_1 \cdot 1\{D_i > d_c\})^2$. This function is clearly continuous in (β_0, β_1) ; it is discontinuous in d_c at $d_c = D_i$. This function is continuous in the parameters of interest with probability one. Suppose that $\|a(\cdot)\|$ is bounded by some function $d(Y_{i,t}, D_i)$ with finite expectation. Then Lemma 2.4 in [Newey and McFadden \(1994\)](#) establishes uniform convergence in probability of $a(\cdot)$. [Wooldridge \(2010, p. 354\)](#) states that the identification condition, along with a uniform weak law of large numbers, establishes consistency of a two-step GMM estimator. $\square$

A.4 Proposition 4

Proof. Observed outcomes are given by $\{Y_{i,\tau}, Y_{i,\tau-1}\}_{i=1}^N$. In period $\tau - 1$, we observe $Y_{i,\tau-1}(D_i, 0)$, which is invariant to dose assignment by the assumption of policy-relevance of the dose (A3). In period τ , we observe $Y_{i,\tau}(D_i, 1)$: under the assumption of a minimum effective dose (A5), $Y_{i,\tau}(D_i, 1) = Y_{i,\tau}(D_i, 0)$ for $D_i < d_c$, and so it follows again by the policy-relevance of the dose (A3) that this variable is also invariant to the dose assignment. $\square$

A.5 Linear Dose Example

From work in the proof to Proposition 4, we have that

$$\begin{aligned}
\widehat{b}_1^{\text{BIN}} &\rightarrow^p E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) | D_i \geq d_r] - E[Y_{i,\tau}(D_i, 1) - Y_{i,\tau}(D_i, 0) | D_i < d_r] \\
&= E[\mu_i(D_i) | D_i \geq d_r] - E[\mu_i(D_i) | D_i < d_r] \\
&= \beta(E[D_i | D_i \geq d_r] - E[D_i | D_i < d_r]) \\
&= \beta \left(\mu + \sigma \frac{\phi(\frac{d_r - \mu}{\sigma})}{1 - \Phi(\frac{d_r - \mu}{\sigma})} - \mu + \sigma \frac{\phi(\frac{d_r - \mu}{\sigma})}{\Phi(\frac{d_r - \mu}{\sigma})} \right) \\
&= \beta \sigma \frac{\phi(\frac{d_r - \mu}{\sigma}) \{ \Phi(\frac{d_r - \mu}{\sigma}) + (1 - \Phi(\frac{d_r - \mu}{\sigma})) \}}{\Phi(\frac{d_r - \mu}{\sigma})(1 - \Phi(\frac{d_r - \mu}{\sigma}))} \\
&= \beta \sigma \frac{\phi(\frac{d_r - \mu}{\sigma})}{\Phi(\frac{d_r - \mu}{\sigma})(1 - \Phi(\frac{d_r - \mu}{\sigma}))}
\end{aligned}$$